

CSE480 Machine Vision

Milestone 1 Report

Jigsaw Puzzle Reconstruction from Scrambled Photographs

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Classical pipeline: enhancement, Canny, CCL, piece description,
weighted shape+colour matching, greedy beam assembly.

1. Introduction

This project reconstructs a jigsaw from a photograph of scrambled pieces. Pieces may appear at unknown 90-degree rotations. The Milestone 1 system is a classical computer-vision pipeline: denoise and stretch the photo, threshold to a binary mask, label connected components, extract contours, describe each piece by four sides (tab / blank / flat plus a colour ribbon), score every legal side pair, and assemble a grid with a greedy beam search.

Rotations are resolved in three layers. Continuous table tilt is removed by deskewing each crop to its min-area rectangle. The remaining 90-degree ambiguity is snapped from observed flats: corner pieces are rotated so flats face north and west in the crop; edge pieces so the single flat faces north. Assembly then searches r in $\{0,1,2,3\}$ (clockwise 90-degree steps). The side that faces board direction D is piece side $(D - r) \bmod 4$, with $D = 0,1,2,3$ for N,E,S,W.

Milestone 2 reuses the same extraction and assembler. Only the compatibility matcher is swapped (Siamese CNN or GNN). This report documents the classical operators, the exact matching formula, assembly tie / dead-end rules, and the limitations of the supplied labels.

2. Problem definition (I/O)

Input: a colour photograph of mixed puzzle pieces on a table, JPEG/PNG, typically 1080p. Optional YOLO label file with the same stem, format ``class cx cy w h`` (normalised), where class is a physical piece identity 1–35.

Output: a reconstructed canvas of an $R \times C$ grid, a placement list (piece id, row, col, rot), intermediate visualisations (enhanced image, mask, labels, contours, Canny stages), and a JSON metrics file containing Q.

Board size is inferred from the extracted piece count: $9 \rightarrow 3 \times 3$, $16 \rightarrow 4 \times 4$, $35 \rightarrow 7 \times 5$. Other counts use the closest factorisation, preferring the configured (7, 5) when the count already equals 35.

Default extraction is connected-component labelling on the enhanced mask (the course spec). YOLO boxes are opt-in (``segmentation.use_gt_boxes: true``) and are otherwise used only to attach piece identities for pose evaluation and ML pair generation.

3. Dataset

The dataset is a Roboflow YOLO export of a 35-piece (7×5) jigsaw photographed in scrambled layouts. Many images show a 9-piece subset; some show 16–21 pieces; a smaller set shows the full 35.

Splits live under `data/input/{train,val,test}` with matching labels under `data/ground_truth/{train,val,test}`. Roboflow augmentations share a source prefix before `.rf.``; the resplit script keeps those variants in one split so train and test never share the same photograph.

Important: YOLO classes are piece identities, not poses. The names list in `data/data.yaml` is not numeric order (`['1','10',..., '9']`). Mapping class index → printed number requires that list. There is no assembled-image ground truth and no rotation label for the photo. Canonical cells assume piece 1 sits at (0,0) row-major on the completed 7×5 board. When a photo's pieces occupy a filled rectangle, that block is compacted (e.g. a 3×3 subset) for pose Q; scattered subsets keep geometry Q only.

4. Methodology overview

Pipeline (shared by classical / Siamese / GNN):

1. Load RGB with Pillow (I/O only).
2. Enhancement: median $3 \times 3 \rightarrow$ Gaussian 5×5 $\sigma=1 \rightarrow$ percentile contrast stretch.
3. Grayscale luminance, Otsu threshold, polarity so pieces are white, morphological close + hole fill.
4. Connected-component labelling from scratch; drop dust by area / aspect / solidity.
5. Moore-neighbour contour on each blob; crop from the original colour image.
6. Corners: convex hull $\rightarrow$ Ramer-Douglas-Peucker $\rightarrow$ largest near-rectangular quad (PCA rectangle fallback). Four sides, signed profile, inward Lab ribbon.
7. Compatibility tensor $D(i,s_i,j,s_j)$. Illegal geometry is $+\infty$.
8. Greedy beam assembly from a top-left corner seed (flats facing N and W).
9. Paste crops onto a canvas; write Q.

All listed filters, Sobel/Prewitt/Canny, CCL, and contours are NumPy implementations. OpenCV is not used for those operators.

5. Enhancement operators

Median ($k=3$). Non-linear order statistic. Each pixel is the median of its 3×3 window (reflect pad). Impulse noise (table specks, JPEG sparkle) is removed without the blur of a mean filter. It cannot be written as a kernel inner product, so the implementation is a justified nested loop (not OpenCV convolution). On this photo set that costs about 17 s per board; that is accepted for the course.

Gaussian ($k=5, \sigma=1$). Separable in principle; implemented as a normalised 2-D kernel

$$G(x,y) = (1/Z) \exp(-(x^2+y^2)/(2\sigma^2)), \quad Z = \sum G.$$

Used both in the enhancement chain and as Canny's pre-smoothing stage.

Contrast stretch. Let v_{low} and v_{high} be the 1st and 99th percentiles of the image.

$$I' = \text{clip}(255(I - v_{\text{low}}) / (v_{\text{high}} - v_{\text{low}}), 0, 255).$$

Percentiles ignore dust min/max so a few bright pixels cannot collapse the range.

Histogram equalisation is implemented (CDF LUT) but not in the default chain: equalising the table photo over-amplifies wood grain and hurts Otsu.

Unsharp mask $I + \alpha(I - G \sigma * I)$ is available for ablation; default α is unused.

Colour strips for matching are taken from `Puzzle.raw_colour` (the unfiltered photo), so photometric matching does not see equalisation or blur.

6. Edge detection and Canny parameters

Sobel kernels produce G_x , G_y through the same ConvolutionEngine as the Gaussian. Magnitude is $\text{hypot}(G_x, G_y)$; orientation is $\text{atan2}(G_y, G_x)$. Prewitt is a drop-in GradientOperator for ablation. Canny then uses the following fixed parameters.

G_σ : Gaussian $k = 5$, $\sigma = 1$

$$|\nabla I| = \sqrt{G_x^2 + G_y^2}, \quad \theta = \text{atan2}(G_y, G_x)$$

NMS: 4 orientation bins $\{0^\circ, 45^\circ, 90^\circ, 135^\circ\}$

$$T_{\text{high}} = P_{90}(\text{NMS} > 0), \quad T_{\text{low}} = 0.4 T_{\text{high}}$$

Hysteresis: 8-connected BFS from strong into weak

Strong pixels are $\text{NMS} \geq T_{\text{high}}$; weak pixels satisfy $T_{\text{low}} \leq \text{NMS} < T_{\text{high}}$. A weak pixel is promoted only if it is 8-adjacent to a strong edge (or to a previously promoted weak pixel). This is the only Canny configuration used in the reconstruction pipeline; $t_{\text{low}} / t_{\text{high}}$ can be injected for unit tests. Contour extraction itself uses Moore tracing on the binary mask, not Canny, because Canny breaks on textured prints. Canny is retained as a required operator and is saved under `results/edge_visualisations/`.

7. Segmentation and piece IDs

Otsu maximises between-class variance $\sigma^2_{B(T)} = w_0(T) w_1(T) (\mu_0 - \mu_1)^2$ on the 256-bin histogram of the enhanced grayscale image.

Polarity: after Otsu, if the majority of pixels (or the brighter region) is labelled foreground, the mask is inverted so the table is background and pieces are white. Without this, a light table becomes one giant blob.

Close ($k=3$) then hole-fill: flood-fill background from the image border; leftover zeros are interior holes (print texture) and are painted foreground.

CCL is a two-pass union-find on 8-connected foreground. Components are rejected if $\text{area} < \max(\text{min_area}, \text{min_area_frac} \times H \times W)$, $\text{bbox aspect} > 5$, $\text{solidity} < 0.35$, $\text{area} < 0.40 \times \text{median blob}$, or $\text{area} > 0.25$ of the frame (table remnant). When YOLO labels exist, keep_n is the unique class count so a 9-piece photo is not assembled on a 2×5 grid of 10 blobs. That uses identities only as a count, not as box crops (the spec path remains CCL).

N = number of surviving components. Grid inference maps the labelled count (when extraction is close) to (R, C) : $9 \rightarrow 3 \times 3$, $16 \rightarrow 4 \times 4$, $35 \rightarrow 7 \times 5$. Extraction IDs are $0 \dots N-1$ in CCL order. Physical piece numbers 1-35 come from YOLO class indices via data.yaml names, attached by box IoU when CCL is used.

8. Piece description and corners

Moore-neighbour tracing walks the boundary of each labelled mask in the crop coordinate system.

Corner method (hybrid, chosen over raw PCA rectangles):

1. Convex hull (Jarvis march) of the contour.
2. Ramer-Douglas-Peucker with $\epsilon = 0.02 \times \text{hull perimeter}$.
3. Among 4-point subsets, keep the quad with large area, similar side lengths, and interior angles in $[55^\circ, 125^\circ]$. Tab tips lie on the hull; a near-rectangular quad prefers the piece body over a tab apex.
4. Fallback: PCA-aligned rectangle snapped back onto the contour, then ordered clockwise from top-left.

Each side is resampled along the contour between consecutive corners. The signed profile is the outward offset from the chord (positive = tab, negative = blank). Classification: mean $|\text{profile}|$ near zero and low variance $\rightarrow$ flat; positive mean $\rightarrow$ tab; negative $\rightarrow$ blank. An inward Lab strip (length 32) is sampled just inside the edge. Out-of-mask samples are NaN, not Lab zeros, so missing colour cannot score as a perfect match.

After sides are classified, the crop is rotated by $k \times 90^\circ$ so flats occupy a canonical local frame (corners: N+W; edges: N). Interior pieces stay in the deskewed frame. Assembly still tries all four rotations.

9. Matching formula and weights

Every ordered pair of sides (i, s_i) and (j, s_j) receives a dissimilarity D . The assembler minimises the sum of D over placed adjacencies. Weights are taken from `configs/classical.yaml` and sum to 1.

$$D = 0.85 E_{\text{shape}} + 0.15 E_{\text{colour}}$$

$$E_{\text{shape}} = \text{mean}((p + \text{flip}(q))^2)$$

$$E_{\text{colour}} = \text{mean} \|Lab_a - \text{flip}(Lab_b)\|_2^2$$

$$D = +\infty \text{ if } i = j, \text{ either side is flat, or both are tab/tab or blank/blank}$$

Profiles are linearly resampled to 64 samples before the shape term; colour strips to 32. Complementary geometry means a tab's positive bump cancels a blank's negative pocket after reversal, so $E_{\text{shape}} \approx 0$ for a true mate. Colour uses only samples that are finite on both sides; if no valid overlap remains, $E_{\text{colour}} = +\infty$ (never 0). Illegal pairs stay IEEE infinity so they cannot win a greedy step. The Siamese matcher converts $p_{\text{neighbour}}$ to a lower-better cost $(-\log p)$ and still marks class-illegal pairs as $+\infty$. GNN uses the same cost convention but is a weak extra on this set (real $\text{val}_{\text{ap}} \sim 0.26$).

10. Assembly: seeds, ties, and dead ends

Algorithm: greedy best-first with beam width $K=8$ (configs/classical.yaml).

Seed. Top-left cell (0,0) is filled by a corner piece: two adjacent flats, rotated so those flats face north and west. If several corners exist, they all enter the beam, ranked by a dummy dissim of 0 then piece id.

Expand. For each beam state, candidate cells are 4-neighbours of already placed pieces. Each unused piece is tried at each of 4 rotations. The placement cost is the sum of D to already-placed neighbours, plus a large finite penalty if a non-flat faces the outer border or a flat faces the interior. Any $+\infty$ neighbour cost discards that move (illegal geometry is never placed).

Ties. After each expansion the beam is sorted by:

1. more pieces placed (primary),
2. lower total dissimilarity,
3. lower minimum piece id as a deterministic last key.

The global best follows the same order.

Dead ends. If no legal growing move exists, the assembler does not force-fill empty cells. It returns the best partial arrangement seen so far. Incomplete boards are expected on hard photos; Q then reflects completeness < 1 .

11. Quality scores

Every run writes Q into `results/evaluation_results/last.json`. There is no assembled-image ground truth, so SSIM is not computed. Lead with identity-neighbour accuracy and complete-reconstruction (usually 0 on this set). Q is the same formula in code, the notebook, and this report. Missing position/orientation terms are 0, so without compact pose GT, $Q \leq 0.2$. Completeness and border-flat accuracy are diagnostics, not Q .

$$Q = 0.5 A_{\text{pos}} + 0.3 A_{\text{ori}} + 0.2 A_{\text{edge}}$$

$$A_{\text{edge}} = A_{\text{idn}} \text{ (identity neighbours)}$$

A_{idn} is the fraction of placed edges whose YOLO piece numbers are 4-adjacent on the completed 7×5 . Orientation A_{ori} is the fraction of border pieces whose placement rot matches the unique identity+flats rot; interiors are omitted (None $\rightarrow$ 0 in Q). If identities compact onto the inferred grid, A_{pos} is reported; otherwise A_{pos} is 0 in Q . `geometry_Q` uses the same formula with tab↔blank accuracy as the edge term (still ≤ 0.2 without pose).

12. Experiments and protocol

Unit tests cover filters, Otsu, Sobel/Canny, CCL polarity, corners, matching illegal= $+\infty$, assembly partial boards, split leakage, and pose-GT mapping (piece 1 $\rightarrow$ (0,0), piece 35 $\rightarrow$ (6,4), compact 3 $\times$ 3).

Reconstruction CLI:

```
python main.py reconstruct --method classical --config configs/classical.yaml \
  --input data/input/test/<file>
```

Methods classical and siamese share extraction and assembly (GNN is an optional ablation). The Siamese matcher fails closed if checkpoints/siamese.pt is missing.

Real-pair training (identity neighbours, relative ori from matching sides):

```
python main.py train-siamese --real --max-boards 80
```

Neighbour identities come from canonical 7 $\times$ 5 adjacency of piece numbers. When both pieces have a unique identity+flats rotation, those rotations map canonical neighbour directions to local sides. Otherwise a unique tab $\leftrightarrow$ blank pair is used, with lowest E_shape as a tie-break (not full classical D). Relative orientation is $(s_i - s_j) \bmod 4$ from those sides — not unlabelled photo rotation. The ori head is trained on that label; inference down-weights pairs whose predicted ori disagrees.

Do not train on data/input/test. After resplit, check_no_source_leakage must pass.

13. Limitations

1. No assembled-image ground truth is provided. Visual inspection of the canvas cannot be replaced by an SSIM to a solved photo. Reported headline metrics are identity-neighbour accuracy and complete reconstruction; Q uses the canonical 0.5/0.3/0.2 formula with missing pose terms as 0.
2. YOLO boxes are identities, not (row, col, rot) in the scrambled image. The canonical 7×5 map (piece 1 at top-left, row-major) is an assumption about how the physical puzzle was numbered; if the box numbering disagrees, pose Q is wrong.
3. Photo rotation is unlabelled. Border placement ori is recovered from identity+flats. Relative ori for ML is $(s_i - s_j) \bmod 4$ from matching sides. Interior piece pose remains unknown. Synthetic complementary ribbons remain a useful pretrain.
4. CCL still fails when pieces touch or when print texture survives hole-fill as bridges. Over-segmentation is reduced by keeping the labelled piece count of largest blobs; YOLO crops avoid residual debris but are not the spec default.
5. Colour is weak: most faces are near-white cardboard, which is why $w_c=0.15$.
6. Greedy beam search is not globally optimal; a wrong early seed cannot be undone.
7. Median is a nested Python order-statistic loop (~17 s/board). Full 35-piece photos remain the expensive case.

Despite these limits, the Milestone 1 operators, formula, weights, Canny parameters, and assembly rules are fully specified and implemented from scratch.

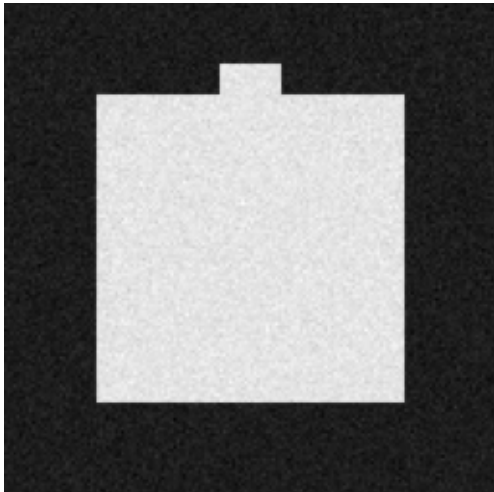
14. Conclusion

Milestone 1 delivers a complete classical reconstruction path with explicit mathematics: $D = 0.85 E_{\text{shape}} + 0.15 E_{\text{colour}}$, illegal pairs at $+\infty$, Canny with Gaussian $k=5$ $\sigma=1$, Sobel, 4-bin NMS, $T_{\text{high}} = 90\text{th percentile of NMS}$, $T_{\text{low}} = 0.4 T_{\text{high}}$, 8-connected hysteresis, and a beam assembler that returns the best partial board on dead ends, breaking ties by fill count then dissim.

Q is numerical on every run: $0.5 A_{\text{pos}} + 0.3 A_{\text{ori}} + 0.2 A_{\text{idn}}$, with missing terms as 0. Do not read a high completeness/border score as a solved puzzle. Milestone 2 plugs Siamese costs into the same tensor slot without forking assembly. GNN remains an optional weak extra.

Enhancement, thresholding, and Canny stages (synthetic piece)

Original



Median $k=3$



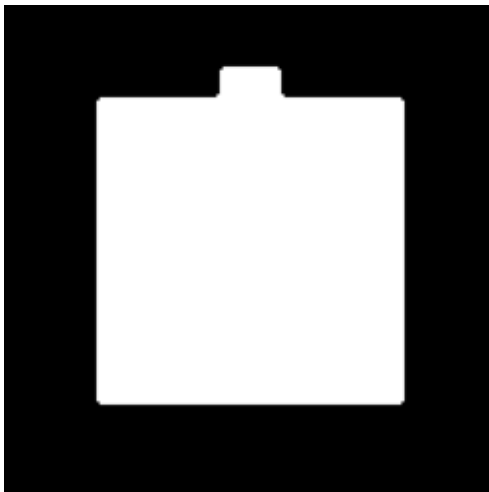
Gaussian $k=5$, $\sigma=1$



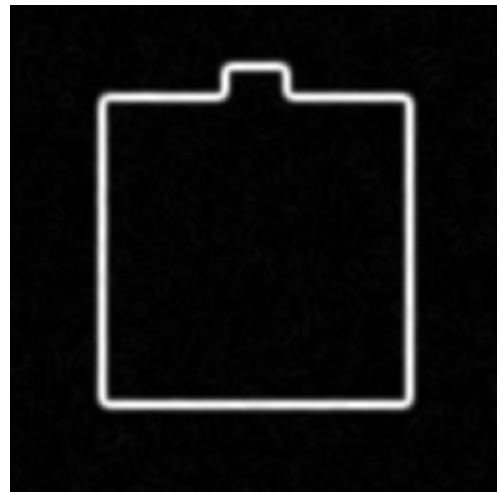
Percentile stretch 1-99



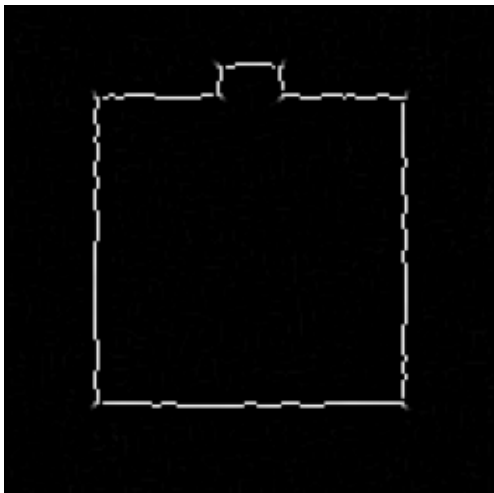
Otsu binary



Sobel magnitude



NMS



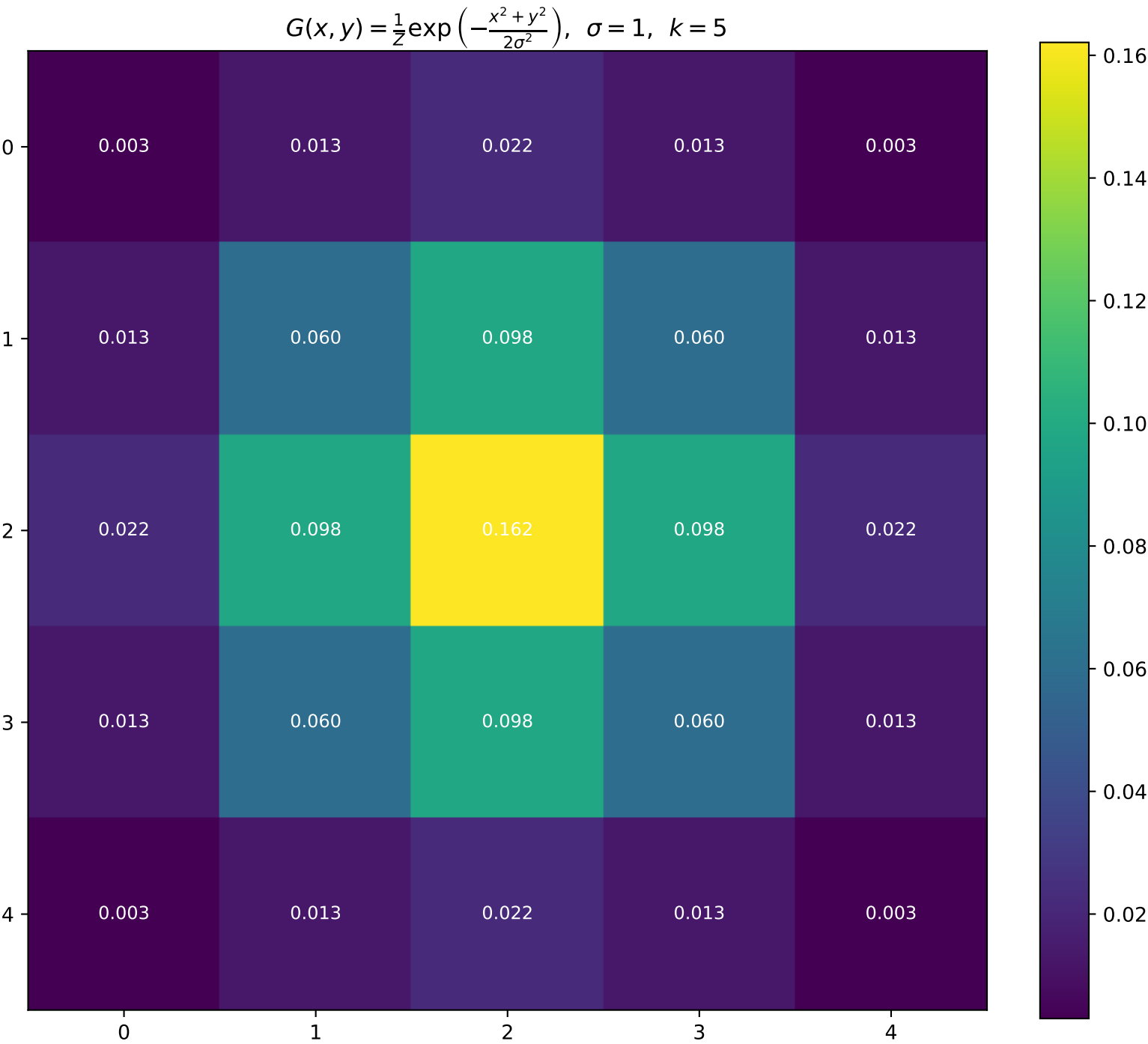
Strong (T_{high})



Hysteresis output

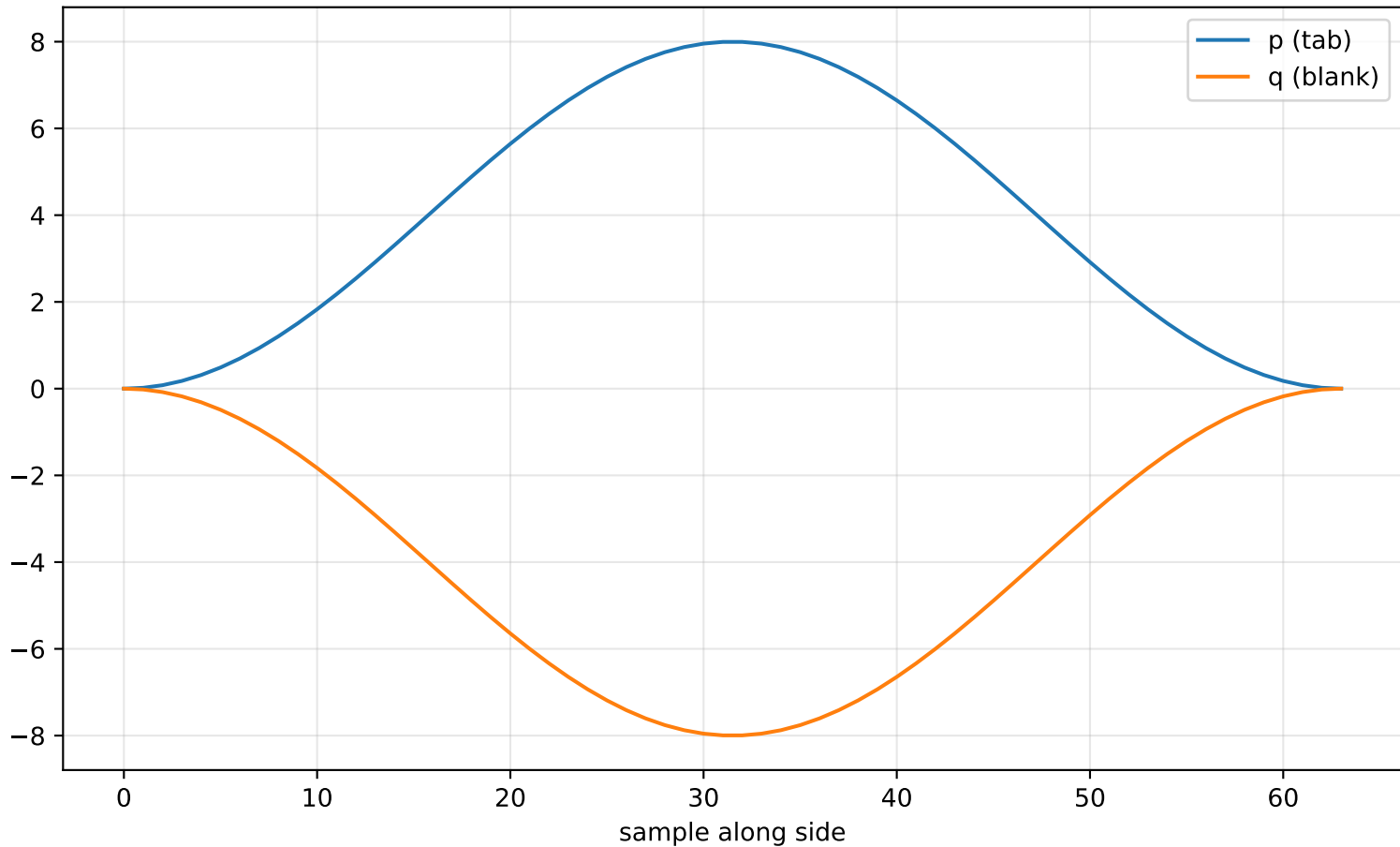


Gaussian smoothing kernel used by Canny (k=5, σ=1)



Complementary shape profiles (tab fills blank)

Raw signed offsets



$p + \text{flip}(q)$ (zero for a perfect mate)

